{COLORIO}

CPU
PI0
PI1
COLORCS
MREAD1
IOCS
MRW
HFLIP
FIXCS
MREAD2
MLWR
MUWR
EXRES
A[1..20]
D[0..15]
DMAINIT
VFLIP
OCHAROMCS
OBJSCRH
OBJSCRV
OBJCS
TIM1
VBLANK
CPUCLK
CPURES
FIXDTAC
MINT16EN
DMACS
V1SCRH
V2SCRH
SVCHAROMCS
VRCSET
SCVRAMCS
SYSWR
MULWR
File: cpu.kicad_sch

Timing
DMACS
FIXCS
HFLIP
VFLIP
RST
P4H
{HVCOUNTERS}
X1S
VCNTEN
D0
CLK2
CLK21
CLK22
CLR
P1H
P2H
N2H
N2HP4H
8H
P256VD
TIM
TIM3
OBJBUFINIT
DMAINIT
DMACLR
OBJBUFCLR
OSCANCLR
FBREF
CSYNC
CBLK
CS1
CS2
RS1
RS2
OE1
OE2
HSYNC
WR
EXRES
VBLANK
CPUCLK
CPURES
FIXDTAC
1VD
SYNCRLDV
TIM1
DMA
RAS
File: timing.kicad_sch

Sprites1
File: sprites1.kicad_sch
CLK21
CLR
P1H
P2H
N2H
N2HP4H
8H
P256VD
TIM
TIM3
OBJBUFINIT
DMAINIT
DMACLR
OBJBUFCLR
OSCANCLR
FBREF
HFLIP
A[1..19]
D[0..15]
MLWR
MUWR
MREAD2
OCHAROMCS
OBJSCRH
OBJSCRV
OBJCS
DMAON
CRTKILL
DMAD
OCHF
FBWM1
FBWM2
VD[0..7]
HD[2..8]
HJ[0..1]
W[0..15]
VPD8
C[0..3]
CX[0..3]

GFX – Fixed layer
CLK2
P1H
P2H
P4H
{HVCOUNTERS}
TIM3
HFLIP
FIXCS
MREAD2
MLWR
MUWR
A[1..11]
D[0..15]
FX[0..7]
File: GFX–Fixed layer.kicad_sch

Sprites 2
CLK21
CLK22
CLR
P2H
N2H
P256VD
X1S
VCNTEN
OCHF
FBWM1
FBWM2
VD[0..7]
HD[2..8]
HJ[0..1]
W[0..15]
VPD8
C[0..3]
CX[0..3]
HFLIP
VFLIP
CS1
CS2
RS1
RS2
OE1
OE2
HSYNC
WR
File: sprites2.kicad_sch

GFX – Tiles
CLK2
EXRES
CLR
P2H
P4H
N2H
P1H
N2HP4H
1V
VCNTEN
SYNCRLDV
TIM1
RAS
HFLIP
HSYNC
V1SCRH
V2SCRH
SVCHAROMCS
VRCSET
SCVRAMCS
SYSWR
MLWR
MUWR
MULWR
MREAD2
A[1..17]
D[0..15]
CRTKILL
1C[0..6]
2C[0..6]
File: GFX–Tiles.kicad_sch

Color mixer
CLK2
A[1..12]
D[0..7]
MLWR
FX[0..7]
OCD[0..7]
1C[0..6]
2C[0..6]
PI0
PI1
COLORCS
MREAD1
CBLK
CRTKILL
REDD
GREEND
BLUED
File: color_mixer.kicad_sch

IO
SPEAKER+
SPEAKER–
BSY
BUSY
RED
GREEN
BLUE
A[1..16]
D[0..7]
IOCS
EXRES
MRW
CSYNC
RST
MINT16EN
DMAON
CRTKILLD
SOUND–ON
SOUND–DATA
SDOD
File: io.kicad_sch

Sound
EXRES
SOUND–ON
SOUND–DATA
SDO
D[0..7]
SPEAKER+D
SPEAKER–D
BSYD
BUSYD
File: sound.kicad_sch

Timing diagrams
File: timing_diagrams.kicad_sch

Sprite timing diagrams
File: sprite_timing_diagrams.kicad_sch

U216D
74LS04
M11

U228B
74LS21
E19

U205E
74LS07
H29

U197C
74LS07
A22

U205D
74LS07
H29

U205F
74LS07
H29

U226C
74LS08
J27

U226B
74LS08
J27

A23F
74LS07

U198A
74LS32
F15

U225B
74LS08
D30

U225A
74LS08
D30

U210A
74LS74
J23

Ulf Skutnabba, twitter: @skutis77
Konami GX870
Sheet: /
File: fround.kicad_sch
Title: The Final Round
Size: A3 | Date: 2024–09–08 | **Rev:**
KiCad E.D.A. 8.0.9 | Id: 1/12

RA2
4.7k x 4
G31
MHALT
MRESET
VCC

RA5
4.7k x 8
E31
MLDS
MUDS
MAS
MRW
MDTACK
MBGACK
MBR
MBERR
VCC

U229
74LS367
H30
CPUCLK
GND
VCC
G31
R48
1k

U222A
74LS74
F20
MINT16END
VBLANK
MBGACK
MBR
IPL0
IPL1
IPL2

U122
68000D
E30
CLK
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
IPL0
IPL1
IPL2
BGACK
BG
BR
A1..A23
D0..D15
AS
UDS
LDS
R/W

U224B
74LS00
G21
H28A
74LS04
MA23
MAS

U228A
74LS21
E19
U205A
74LS07
H29
U205C
74LS07
U205B
CPURES

U227B
74LS74
E20
U222B
74LS74
SCVRAMCS
TIM1
U204B
H27
U206E
F19

VOROMDTAC
FIXDTAC
J27
U226D
74LS08

MAS
A20
DMAINIT
OBJCS
U191A
G24
U220C
J21
XCS
U191D
G24

U192B
74LS32
F21
MUWR
MLWR
U192A
74LS32
F21
U211A
74LS08
L25
MULWR

MA[1..14]
A[1..20]
MD[0..15]

U88
27C010
F27
U139
G25
MUWR
WORKCS
6264SL

U137
27C010
E27
U105
E25
MUWR
WORKCS
6264SL

74LS367
U113
H26
74LS245
U107
F24
U96
F23
U141
E24
buffers
A->B
MREAD1

A[1..20]
D[0..15]

U224A
74LS00
G21
ROMCS
WORKCS

FIXCS
SCVRAMCS
OBJCS
0x10 0000.. 0x1F FFFF
0x12 0000.. 0x13 FFFF
0x14 0000.. 0x15 FFFF
Fixed layer RAM
Tile map 1 and 2 RAM
Sprite RAM

U223B
74LS139
H23
decoder
H28F
74LS04
U189A
H24

U206B
74LS04
F19
U206C
74LS04
F19
MREAD1
MREAD2

U188A
74LS32
K25
DMACS

U128
74LS138
address decoder
G23
U226A
74LS08
J27
ROMCS
WORKCS
COLORCS
IOCS
VRCSET
SYSWR

U3
74LS138
K26
OBJSCRH
OBJSCRV
V1SCRH
V2SCRH
SYSWR
MULWR

U89
74LS273
O19
flip flops
VFLIP
HFLIP
PI0
PI1
EXRES

U223A
74LS139
H23
decoder
H28B
74LS04
U191B
G24
XCS
SVCHAROMCS
OCHAROMCS

U209A
H21
U209B
H21
U211C
L25
74LS08
CPUCLK
74LS74
VOROMDTAC

ROMCS
WORKCS
COLORCS
IOCS
SYSWR
VRCSET
0x0 0000.. 0x3 FFFF
0x4 0000.. 0x5 FFFF
0x8 0000.. 0x9 FFFF
0xA 0000.. 0xB FFFF
0xC 0000.. 0xD FFFF
0xE 0000.. 0xF FFFF
ROM
Communication RAM
Palette
System I/O
System registers
(It can also be used for read) DMACS is activated)
Graphics bank

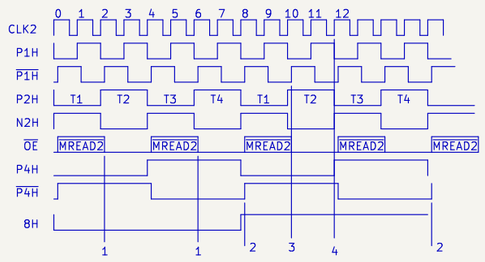
XCS
SVCHAROMCS
OCHAROMCS
0x00 0000..0x1F FFFF
0x40 0000..0x5F FFFF
0x60 0000..0x7F FFFF
Tile ROM
GFX ROM

CPU
Konami GX870
Sheet: /CPU/
File: cpu.kicad_sch
Title: The Final Round
Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 8.0.9 | Id: 2/12

the four bits of each layer has the following meaning:
0000: Transparent
1111: Shadow
Else: Opaque

Color Mixer	
Konami GX870	
Sheet: /Color mixer/	
File: color_mixer.kicad_sch	
Title: The Final Round	
Size: A3	Date: 2024-09-08
KiCad E.D.A. 8.0.9	Id: 3/12

CLK2: 6.1397MHz
1V: 8.00kHz
2V: 4.00kHz
4V: 2.000kHz
P1H: 3.0719MHz
P2H: 1.536MHz
P4H: 0.767971MHz
8H: 384.000kHz
16H: 192.000kHz



RAM Control

Slot	Control
T1,T3	CPU
T2,T4	Video circuit

Time

1	Data output from RAM to Video circuit
2	Data latched into first DFF's
3	Data latched into second DFF's.
4	Pixel data output to FX[0..7]

The delay from the LSB (8H) of the RAM accessed by the video circuit is six pixels to output.

8-bits of data is latched every P4H cycle.
It is 2 pixels = 8bits / (4 bits / pixel).
The current pixel is multiplexed with P1H (also HFLIP).

B0 goes to pin 36 on timing circuit

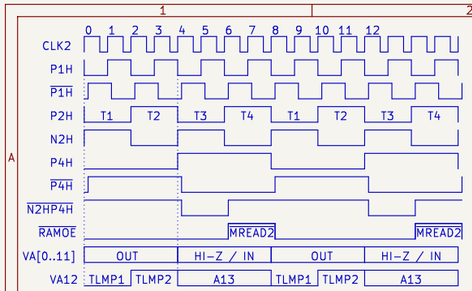
Kolla 8V och 256H

When P2H is high I1 is selected.

FXD[0..8]: Tile number
FXD[9..12]: Palette color
FXD[13]: HFLIP, not used on Final round
FXD[14]: VFLIP, not used on Final round

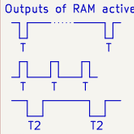
RAM
A[0..5] Horizontal position 0..511
A[6..10] Vertical position 0..255 in steps of 8
A[11.12] Tilemaps? Up to 4.

Fixed Layer
Konami GX870
Sheet: /GFX - Fixed layer/
File: GFX-Fixed layer.kicad_sch
Title: The Final Round
Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 8.0.9 | Id: 4/12



When VA is switched to output the bus is updated for each P1H cycle.

Outputs of RAM active when P4H is low, i.e during T1 and T2.



SYNCRLDV: Usually 60.604Hz
T is ~150ns.

VCNTEN: 8.0000kHz

HSYNC: 16.0000kHz, T2: 5.2175us

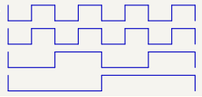
Tilemap 1 and 2
VA[5..0]: Horizontal position: 0..63 -> 0..63 x 8 = 0..504
VA[11..6]: Vertical position: 0..63 -> 0..63 x 8 = 0..504
VA12 = 0: Tilemap 1, VA12 = 1: Tilemap 2

It is unclear what SYNCRLDV (Synchronized Reload Vertical?) and VCNTEN (Vertical Counter Enable?) are used for. They become active before HSYNC goes low. They might be used to synronize the output of the vertical counters so they are updated after a write to scroll registers before HSYNC goes low, and the vertical counters.

Data latched from the RAM:

Edge	Tilemap
T1-T2	1
T2-T3	2

It is output 2 pixel clocks later to the ROMs.



The vertical counters are not the same value when they are offset with the scroll registers.

U116 74LS157

U188B 74LS32 K25

74LS04 M11 U216A

74LS245 U31 K27 buffers

74LS245 U33 L27 buffers

74LS374 U134 M25 3-state flip flops

74LS374 U132 L26 3-state flip flops

74LS374 U81 M27 3-state flip flops

74LS374 U79 M28 3-state flip flops

74LS374 U87 M26 3-state flip flops

74LS374 U92 M29 3-state flip flops

74LS32 U195B H25

74LS32 U188C K25

74LS32 U218A L23

74LS32 U218C L23

74LS32 U218D L23

74LS32 U189C H24

74LS32 U188D K25

74LS08 U211B L25

74LS08 U211D L25

74LS157 U140

74LS157 U112

74LS157 U27

74LS157 U28

74LS157 U93

74LS257 U110 P25

74LS257 U114 Q25

74LS257 U91 Q24

74LS257 U83 P24

U136 MX23C4100

U138 MX23C4100

U36 6264SL M21

U38 6264SL M23

J30 U34 007779

U45 007781 O22

74LS174 U52 M24

74LS174 U54 O24

A[1..17]
D[0..15]
VA[0..11]
{V}
E[0..12]
V[0..15]
EV[1..4]
CA[0..17]
C[0..31]
B[20..27]
2C[0..6]
1C[0..6]
2S[0..2]
1S[0..2]

MREAD2 MULWRD VRCSET M11 N2HP4HD SCVRAMCSD P2HD P4HD RAS MUWRD VRU SCVRAMCS TIM1 MLWRD A13 RAMOE MREAD2D

VCC GND

SYNCRLDV CLK2 P1HD HSYNCD VCNTEND 1VD CLRD MULWRD SYSWRD

SAB12 SAB11 SAB10 SAB9 SAB8 SAB7 SAB6 SAB5 SAB4 SAB3 SAB2 SAB1 SYLD CLK2 P1H HSYNC VCEN E1V CLR MULW SFLG

DB8 DB7 DB6 DB5 DB4 DB3 DB2 DB1 DB0 E4V2 E2V2 E1V2 E4V1 E2V1 E1V1

A12 A11 A10 A9 A8 A7 A6 A5 A4 A3 A2 A1 AB12 AB11 AB10 AB9 AB8 AB7 AB6 AB5 AB4 AB3 AB2 AB1 VA11 VA10 VA9 VA8 VA7 VA6 VA5 VA4 VA3 VA2 VA1 VA0

CA0 CA1 CA2 CA3 CA4 CA5 CA6 CA7 CA8 CA9 CA10 CA11 CA12 CA13 CA14 CA15 CA16 CA17 D15/A-1

C31 C30 C29 C28 C27 C26 C25 C24 C23 C22 C21 C20 C19 C18 C17 C16 C15 C14 C13 C12 C11 C10 C9 C8 C7 C6 C5 C4 C3 C2 C1 C0

V15 V14 V13 1S2 1S1 1S0 2S2 2S1 2S0 CLR CLK2 HFLIP

VC31 VC30 VC29 VC28 VC27 VC26 VC25 VC24 VC23 VC22 VC21 VC20 VC19 VC18 VC17 VC16 VC15 VC14 VC13 VC12 VC11 VC10 VC9 VC8 VC7 VC6 VC5 VC4 VC3 VC2 VC1 VC0 V2C0 V2C1 V2C2 V2C3 V2C4 V2C5 V2C6 V1C0 V1C1 V1C2 V1C3 V1C4 V1C5 V1C6 VD15 VD14 VD13 V1S2 V1S1 V1S0 V2S2 V2S1 V2S0 TES1 TES2

D0 D1 D2 D3 D4 D5 D6 D7 D8 D9 D10 D11 D12 D13 D14 D15

B20 B21 B22 B23 B24 B25 B26 B27

CA14 CA15 CA17 CA16 CA12 CA13 CA8 CA9 CA11 CA10 CA4 CA5 CA7 CA6 CA0 CA1 CA3 CA2

D12 D13 D15 D14 D8 D9 D11 D10 D4 D5 D7 D6 D0 D1 D3 D2

1S0 1S1 1S2 2S0 2S1 2S2

V8 V9 V10 V11 V12 V13 V14 V15 V0 V1 V2 V3 V4 V5 V6 V7

E5 E6 E7 E8 E9 E10 E11 E12 EV1 EV2 EV4 E0 E1 E2 E3 E4

1V1 2V1 4V1 V0 V1 V2 V3 V4 1V2 2V2 4V2

Address Register
0x06 V1SCRH Tilemap 1, horizontal scroll
0x08 V1SCRV Tilemap 1, vertical scroll
0x10 V2SCRH Tilemap 2, horizontal scroll
0x12 V2SCRV Tilemap 2, vertical scroll

V1SCRV and V2SCRV are in 007779 and are accessed when SYSWR and MULWR are active.

A total of 9-bits are used for vertical scroll.
2^9 = 512 scroll positions.

A13 = Hi: VRAM2(Tilemap 2).
A13 = Low: VRAM1(Tilemap 1)

V[13..15]: Color
V[11,12]: Tile bank
V[0..10]: Tile number

The 007781 is a line buffer. All counters related to all timing signals are inside the chip. It is synchronized on the rising edge of CLR and CLK2 after that.

1C[0:3]: Layer 1 Pixel color code
1C[4:6]: Layer 1 Palette
2C[0:3]: Layer 2 Pixel color code
2C[4:6]: Layer 2 Palette

Tile Layers
Konami GX870
Sheet: /GFX - Tiles/
File: GFX-Tiles.kicad_sch

Title: The Final Round

Size: A3	Date: 2024-09-08	**Rev:**
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According to mame the object table
for each frame buffer seems to be
size 1800h = 6144 words. That is after
processing 6144 / 80 = 76 sprites

Object Color Data

BEFLAG = Bus Enable FLAG?

OCHF: horizontal flip

{ULRW}

{DMARW}

007783
When BEFLAG goes active it will
use the low byte OD[7..0] to create the
address for the writes to sprite RAM.
According to mame:
OA[12..0] = 0x1800 | ((OD[7..0] & 0xff) << 2)

007784
– subtracts the scroll value for the
sprites from current coordinates for the sprites.
– When DMA goes high. The 007784 will read 8 words, if bit OD15
of the first word is high then BEFLAG goes high and stays high
until the sprite RAM has processed the first four words and possibly
updated the four following words at the high end of the RAM.
it will compress the eigth words into four
words and also subtract the x– and y–scroll values from the x and y
coordinates. it will write the four words at the high end of the
sprite RAM.
– Addresses at the low end goes from 0x0000..0x0007 and the following
cycle at 0x0028..0x002F..

Scroll Register
OBJSCRV Sprite horizontal scroll
OBJSCRH Sprite vertical scroll

Data is latched in the scroll registers when
the signals OBJSCRV and OBJSCRH go low.

The sprite memory is controlled in three separate cases:
1. OBJBUFINIT
2. BEFLAG
3. ~DMAINIT, when OBJBUFINIT and DMA are low.

007793

CLK21′ = CLK21

$\overline{LRW}$ = ~($\overline{DMAONS}$ & OBJBUFINIT & $\overline{CLK21'}$ +
BEFLAG & DMA & P1H & $\overline{DMAONS}$ & CLK21′ & (~8H & HFLIP + 8H & ~HFLIP)
~$\overline{LRW}$ & ~DMAINIT)

$\overline{URW}$ = ~($\overline{DMAONS}$ & OBJBUFINIT & $\overline{CLK21'}$ +
BEFLAG & DMA & P1H & $\overline{DMAONS}$ & CLK21′ & (8H & ~HFLIP + ~8H & HFLIP) +
~$\overline{URW}$ & ~DMAINIT)

007785

HJ[0..1] Unknown

C[0..3] Palette value for frame buffer 1
CX[0..3] Palette value for frame buffer 2

O[17:19]: RAM and ROM bank selects. Not used
on final round.

If VPD8 is high, the vertical position
is outside of visible area. The frame
buffer is not updated then.

Sprite Layer – Part 1
Konami GX870
Sheet: /Sprites1/
File: sprites1.kicad_sch

Title: The Final Round

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 6/12

Pixel palette data

Frame Buffer 1

Frame Buffer 2

Pixel color data

Framebuffer 1 is outputting pixel data when P256VD is low and writes to the FB is possible when P256VD is high. The conditions are reveresed for FB2.

$\overline{FBWM1}$: usually f= 250Hz, sometimes 1.15kHz
$\overline{FBWM2}$: Mostly active all the time. When it is not active f = 60.06Hz with small active pulses, not synced to $\overline{FBWM1}$. It becomes non active 25 horizontal lines after VBLANK goes active.

Writes to FB are possible when the "pixel part" is sleected from the mux.
If this expression is true
(R[0:3] != 4'h0) && ($\overline{FBWM1}$ | $\overline{FBWM2}$ | ~$\overline{WR}$ | ~VPD8)
data is written to DRAM.

Data is written when:
– $\overline{WR}$ is active.
– Not a transparent pixel (4'h0).
– $\overline{FBWM1}$ and $\overline{FBWM2}$ are not active: No writes to frame buffer going on.
– VPD8 (vertical counter highest bit, VCNT > 256) is not active. Outside visible area.

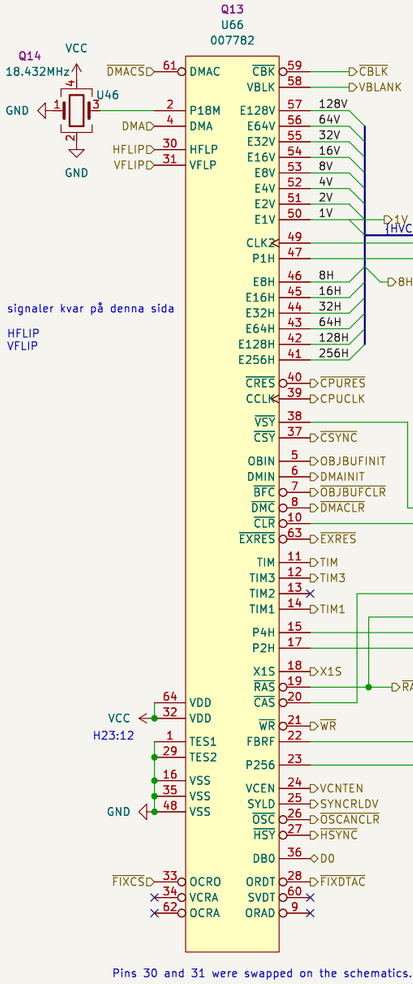
007787

Multiplexer and buffer.
– When P256VD is low R[0:27] are the active inputs. When P256VD is high r[0:27]X are the active inputs.
– It will latch 4 nibbles (R[0:3], R[8:11],..) at a time and four CLK2 cycles later output one nibble at a time at the pixel clock rate.

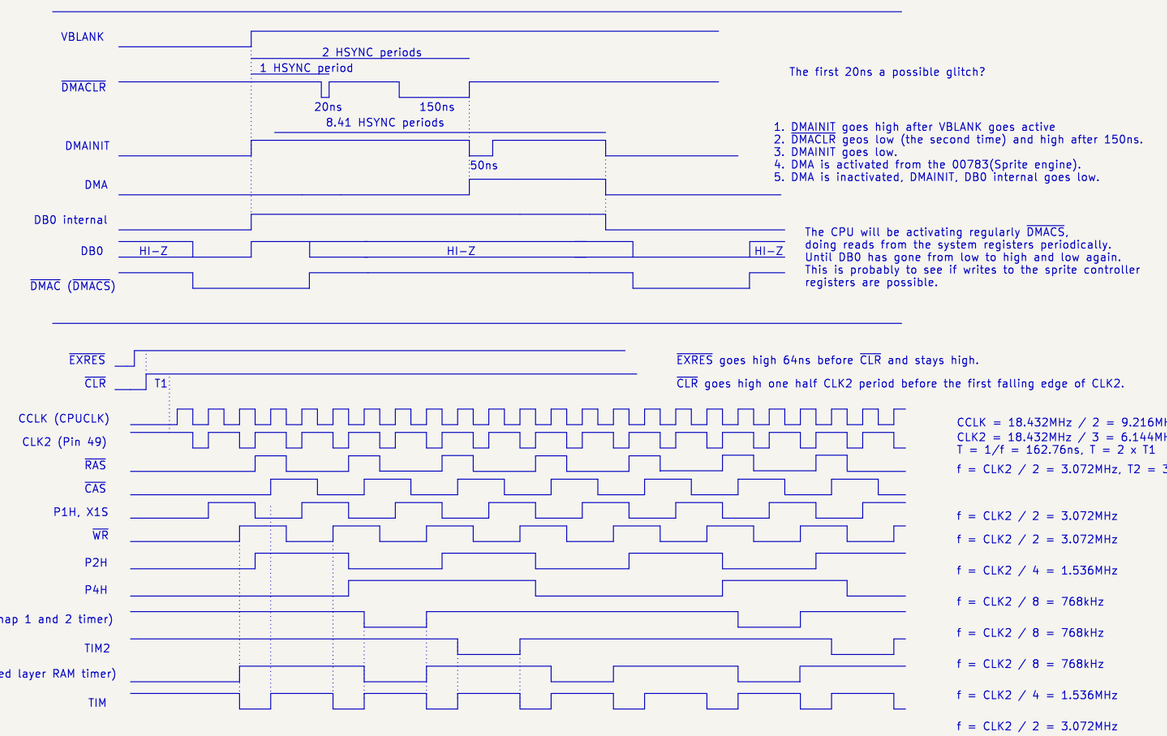
Sprite Layer – Part 2
Konami GX870
Sheet: /Sprites 2/
File: sprites2.kicad_sch

Title: The Final Round

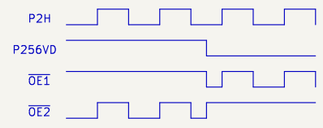
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 7/12



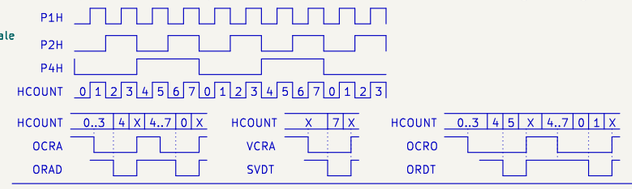
DMA Synchronization



Output enables active vs. part of frame buffer active



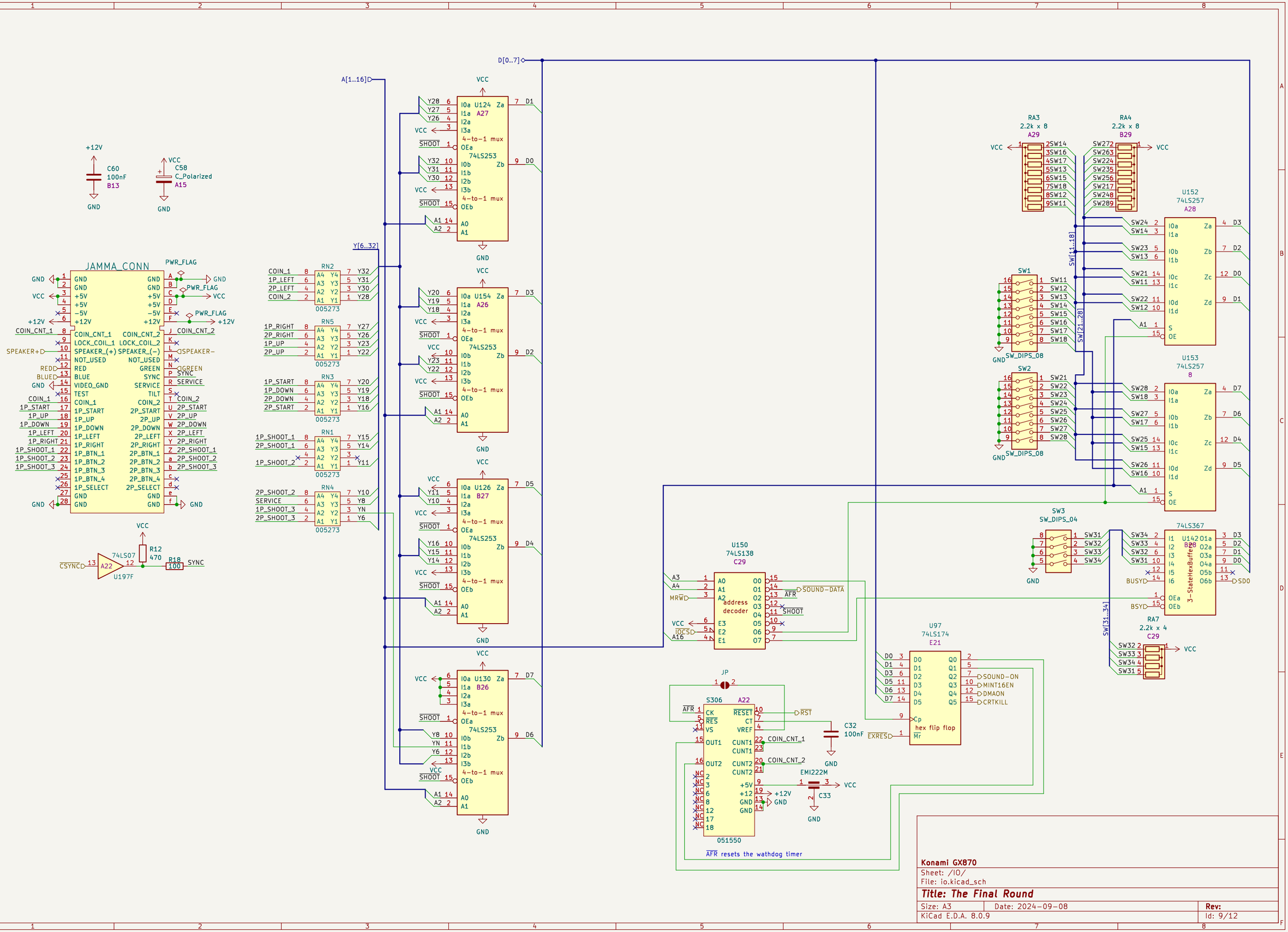
Chip select signals and Data Acknowledge signals

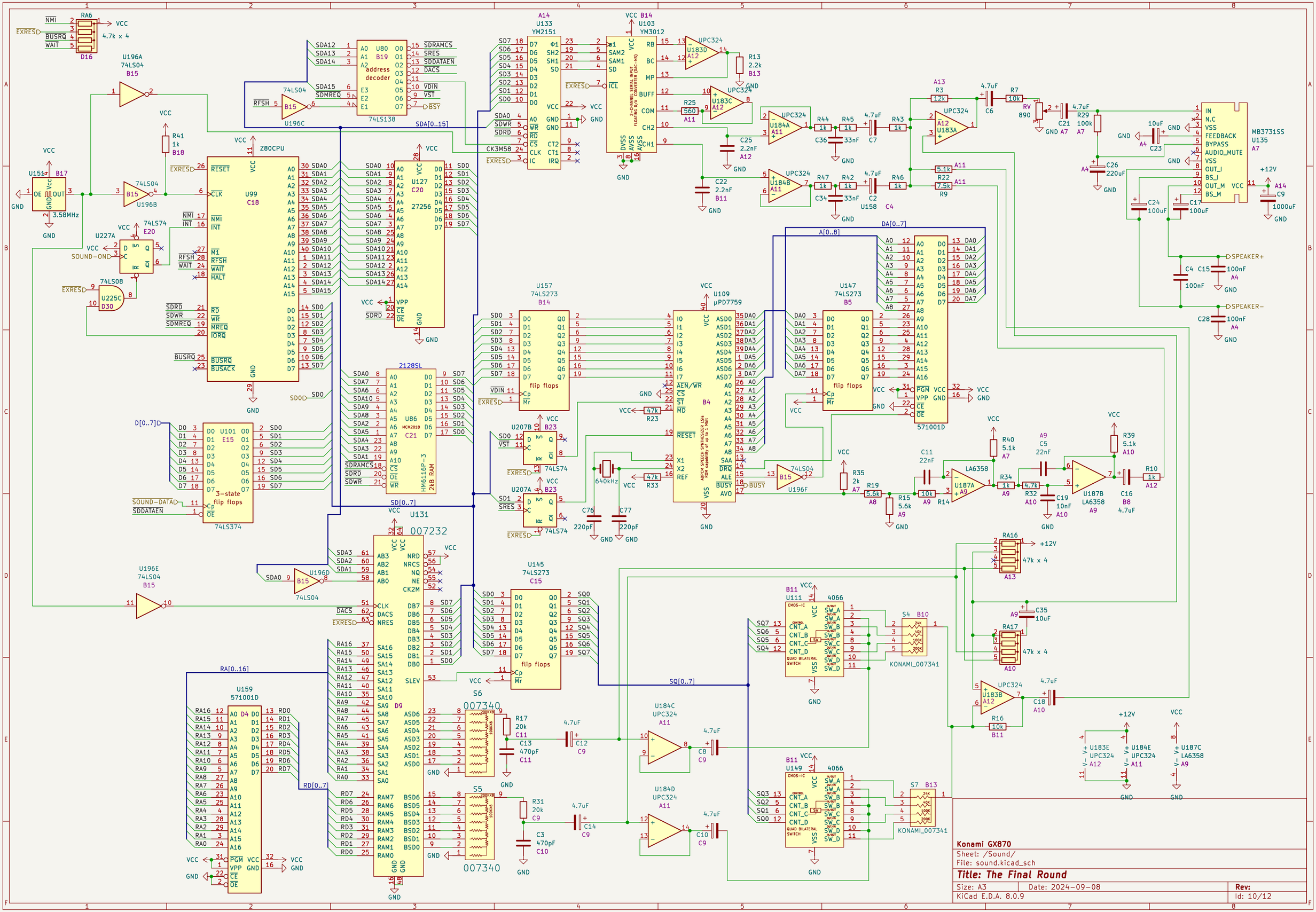


Timing
Konami GX870
Sheet: /Timing/
File: timing.kicad_sch

Title: The Final Round

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D[0..7]
A[1..16]
Y[6..32]
JAMMA_CONN
PWR_FLAG
C60
100nF
B13
C58
C_Polarized
A15
RN1
RN2
RN3
RN4
RN5
005273
U124 A27
U154 A26
U126 B27
U130 B26
74LS253
4-to-1 mux
U150
74LS138
C29
address decoder
U97
74LS174
E21
hex flip flop
U152
74LS257
A28
U153
B8
RA3
RA4
2.2k x 8
A29
B29
SW1
SW2
SW3
SW_DIPS_08
SW_DIPS_04
U142
74LS367
B28
3-StateHexBuffer
RA7
C29
S306
A22
051550
U197F
74LS07
A22
R12
470
R18
100
JP
C32
100nF
EMI222M
C33
AFR resets the wathdog timer
Konami GX870
Sheet: /IO/
File: io.kicad_sch
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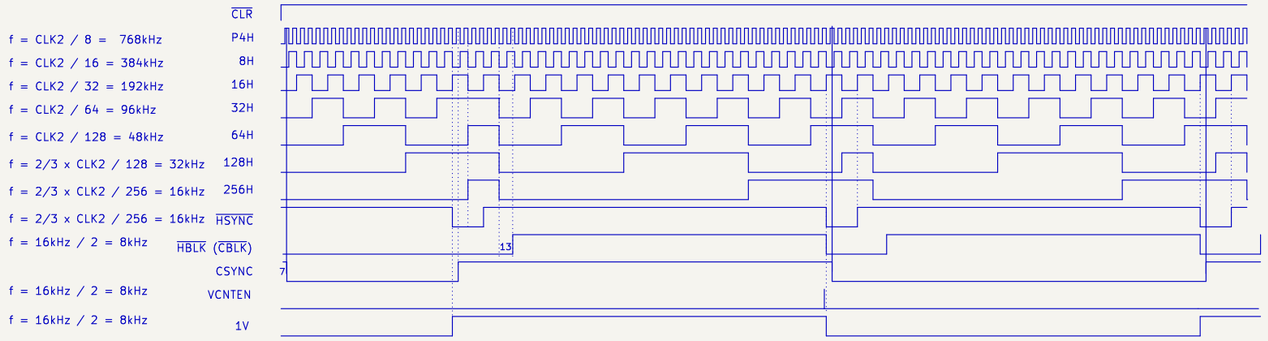
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Sheet: /Sound/
File: sound.kicad_sch

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Horizontal signals

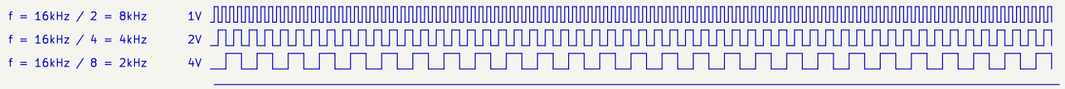


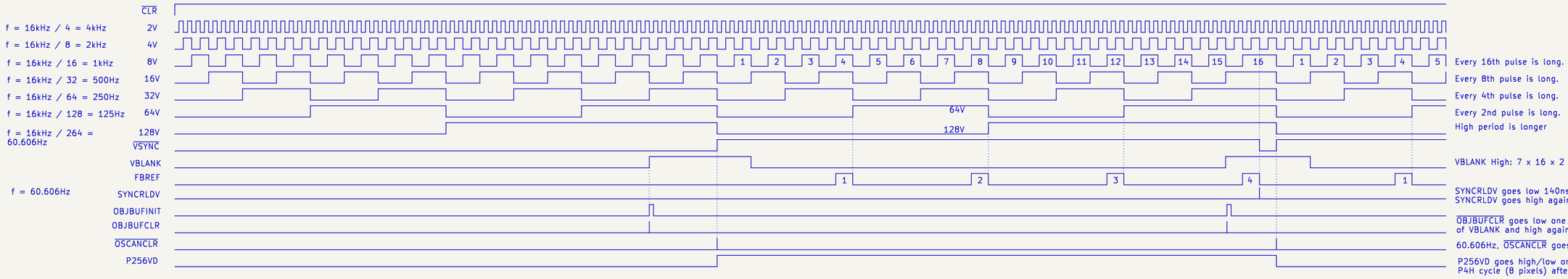
– The first VCNTEN is skipped after reset.
It goes low 140ns before HSYNC goes low,
and high again when HSYNC goes low.
VCNTEN is active right before every second falling edge of HSYNC.

– CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

The first HSYNC is at HCOUNT 175 after reset.

Vertical signals





FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

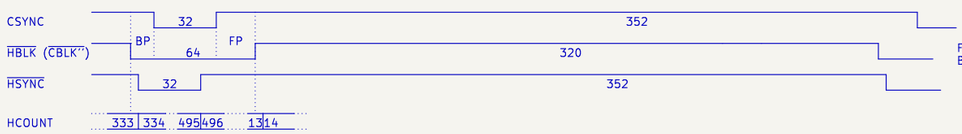
Horizontal and vertical synch timing diagrams

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.

Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

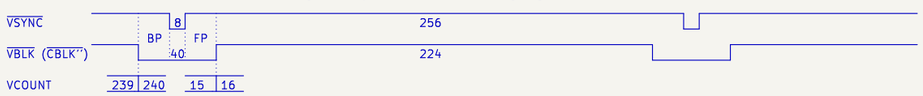
Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40



Front Porch (FP) 24 px,
Back Porch (BP) 8 px

The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



Front Porch (FP) and
Back Porch (BP) 16 HSYNC cycles.

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

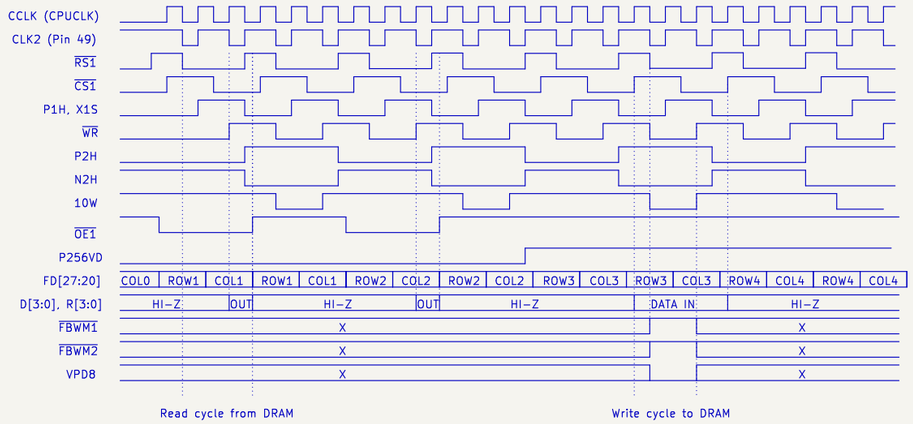
CBLK'' is at the output of color mixer.

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Sheet: /Timing diagrams/
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The same row and column addresses are repeated twice.
One for read cycle and the other for write cycles.

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Sheet: /Sprite timing diagrams/
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